

Madhur Panwar

Ph.D. Student, EPFL

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EDUCATION

- **École Polytechnique Fédérale de Lausanne (EPFL)** Lausanne, Switzerland
Ph.D. in Computer and Communication Sciences 2024 – Present
 - Advisor: Antoine Bosselut (NLP Lab)
- **Birla Institute of Technology and Science, Pilani (BITS Pilani)** Pilani, India
B.E. Computer Science and M.Sc. Mathematics (Dual Degree) 2016 – 2021
 - CGPA: 9.32/10.00 (**Distinction division**)
 - Recipient of Pratibhashali Award (Department of Mathematics)

EXPERIENCE

- **Arcadia Impact** Remote
Research Engineer & Technical Project Manager July 2024 – March 2025
 - Worked part-time under the [AI Safety Engineering Taskforce \(ASET\)](#) to build and contribute AI safety evaluations to leading LLM evaluation frameworks: UK AISI's Inspect and METR's Vivaria.
 - Led and mentored a team of engineers developing agentic AI safety evaluations under the [UK AISI bounty programme](#).
- **Microsoft Research** Bangalore, India
Research Fellow – Advisor: [Dr. Navin Goyal](#) Aug 2022 – Aug 2024
 - Worked on understanding the emergent abilities of large language models and how they represent natural language. Our work on In-Context Learning and Bayesian Inference was **accepted at ICLR 2024** and **various NeurIPS 2023 Workshops**.
- **Adobe** Noida, India
Software Development Engineer – Manager: [Ganesh R](#) Jul 2021 – Jul 2022
 - Developed the Landing Pages feature in [Adobe Journey Optimizer \(AJO\)](#), allowing marketers to design and serve landing pages to millions of end users. Received Spot Award for fixing a Customer Service Outage and won the New Hire Coding Bootcamp. Mentored summer internship projects for three students across Machine Learning and Software Engineering domains.
- **Nanyang Technological University** Singapore
Undergraduate Thesis – Advisors: [Prof. Chng Eng Siong](#), [Prof. Poonam Goyal](#) Jan 2021 – Jun 2021
 - Developed a fully Pythonic pipeline for speaker diarization using the x-vector representations and LSTM-based spectral clustering. [[thesis](#)]
- **Amazon** Hyderabad, India
Software Development Engineer Intern – Manager: [Pankaj Jain](#) Aug 2020 – Dec 2020
 - Developed a new module end-to-end for generating the node launch configuration of a major Amazon carrier. The developed features were deployed to production.
- **Adobe** Noida, India
Research Intern – Advisors: [Balaji Krishnamurthy](#), [Milan Aggarwal](#) May 2020 – Jul 2020
 - Worked at the Media and Data Science Research (MDSR) Lab and developed novel approaches to topic modeling. This research has been published as [a long paper at ACL 2021 \(Oral\)](#).
- **University of Victoria** British Columbia, Canada
Mitacs Globalink Research Intern – Advisor: [Prof. Ralph Evins](#) May 2019 – Aug 2019
 - Worked on the development of an open-source project, [BESOS](#), that has been [published in The Journal of Open Source Software \(JOSS\)](#). Formulated and solved building energy optimization problems.
- **Indian Institute of Remote Sensing, Indian Space Research Organisation** Dehradun, India
Research Intern – Advisor: [Dr. Shashi Kumar](#) May 2018 – Jul 2018
 - Developed a system to calibrate Synthetic Aperture Radar (SAR) satellite imagery and a plugin to detect oil spills in SAR images.

- [1] **For Better or for Worse, Transformers Seek Patterns for Memorization** [pdf]
Madhur Panwar, Gail Weiss, Navin Goyal, Antoine Bosselut
The Thirty-ninth Annual Conference on Neural Information Processing Systems [NeurIPS '25]
- [2] **Learning Syntax Without Planting Trees: Understanding Hierarchical Generalization in Transformers** [pdf]
 Kabir Ahuja, Vidhisha Balachandran, Madhur Panwar, Tianxing He, Noah A. Smith, Navin Goyal, Yulia Tsvetkov
Transactions of the Association for Computational Linguistics [TACL '25]
ICML 2024 Workshop on Mechanistic Interpretability (Spotlight)
- [3] **InversionView: A General-Purpose Method for Reading Information from Neural Activations** [pdf]
 Xinting Huang, Madhur Panwar, Navin Goyal, Michael Hahn
The Thirty-eighth Annual Conference on Neural Information Processing Systems [NeurIPS '24]
ICML 2024 Workshop on Mechanistic Interpretability [Oral; top paper - 2nd place (500 USD)]
- [4] **In-Context Learning through the Bayesian Prism** [pdf]
Madhur Panwar^{*}, Kabir Ahuja^{*}, Navin Goyal
The Twelfth International Conference on Learning Representations [ICLR '24]
- [5] **TAN-NTM: Topic Attention Networks for Neural Topic Modeling** [pdf]
Madhur Panwar^{*}, Shashank Shailabh^{*}, Milan Aggarwal^{*}, Balaji Krishnamurthy
In Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics [ACL '21 (Oral)]
- [6] **In-Context Learning and Bayesian Inference** [pdf]
Madhur Panwar^{*}, Kabir Ahuja^{*}, Navin Goyal
NeurIPS 2023 Workshop - Robustness of Few-shot and Zero-shot Learning in Large Foundation Models (R0-FoMo)
- [7] **Surprising Deviations from Bayesian View in In-Context Learning** [pdf]
Madhur Panwar, Kabir Ahuja, Navin Goyal
NeurIPS 2023 Workshop - I Can't Believe It's Not Better (ICBINB): Failure Modes in the Age of Foundation Models
- [8] **Transformers Can Learn To Solve Linear-Inverse Problems In-Context** [pdf]
 Kabir Ahuja^{*}, Madhur Panwar^{*}, Navin Goyal
NeurIPS 2023 Workshop - Deep Learning and Inverse Problems
- [9] **[Re] AdaBelief Optimizer: Adapting Stepsizes by the Belief in Observed Gradients** [pdf]
 Anirudh Buvanesh^{*}, Madhur Panwar^{*}
ReScience C, vol. 8, no. 2, 2022 (accepted under *ML Reproducibility Challenge 2021*) [ReScience C'22]
Presented at NeurIPS 2022 Journal Track (Spotlight)

PATENTS

- [1] **System and Methods for Neural Topic Modeling using Topic Attention Networks** [pdf]
 Shashank Shailabh, Madhur Panwar, Milan Aggarwal, Pinkesh Badjatiya, Simra Shahid, Nikaash Puri, S Sejal Naidu, Sharat Chandra Racha, Balaji Krishnamurthy, Ganesh Palwe
US Patent Application No. 17/644,856 | Adobe Inc.

ACHIEVEMENTS, HONOURS AND AWARDS

- **Special Mention for Best Presentation, Lemanic Life Sciences Hackathon** [🏆], **2026** | 14 teams (> 130 participants) from across the EU participated. We developed an app to visualize complex 3D bird flight trajectories from tracking data, supporting behavioral prediction and infrastructure planning.
- **Runner-up, EuroTeQaThon** [🏆], **2025** | We were second in our group. 24 student teams from 9 top European universities presented their solutions at the Eindhoven University of Technology (TU/e). We (team Treeple) developed a tech-based solution that suggests optimal tree planting for maximum benefits.
- **Winner, 1st place (2,000 CHF), Lemanic Life Sciences Hackathon** [🏆], **2025** | 12 teams (> 110 participants) from across the EU participated. We developed an automated AI-driven system to assign comprehensive rating to games on app playstores, which serves as a tool for early detection of problem gambling (or ludopathy) in children.
- **Research Week with Google, 2023** | Selected by Google Research India among nationwide applicants to attend the "Research Week with Google" event. [certificate]
- **Institute Merit Scholarship, 2016 - 2021** | Received scholarship at BITS Pilani for being in the top 4% of students across all academic departments for all 10 semesters.

- **Hacktoberfest** [🔗], **2021** | Contributed to open source projects like [microsoft/ML-For-Beginners](#), [pytorch-ignite](#), [kedro](#) & [TheAlgorithms/Java](#), and was one among the first 50,000 participants to complete the Hacktoberfest challenge.
- **Pratibhashali Award, 2020** | Conferred by the Department of Mathematics, BITS Pilani, for notable achievements in research and academics. [[certificate](#)]
- **Mitacs Globalink Research Internship (GRI)** [🔗], **2019** | I was one among the 64 Indian applicants selected for this program by [Mitacs](#) in consultation with AICTE [[AICTE report, page 47](#)]. Under GRI, Mitacs provides funding to conduct research in Canada. This competitive program involves international undergraduate applicants from 15 countries.
- **INSPIRE Scholarship** [🔗], **2015** | Awarded by [DST](#), Government of India, for securing a place among the **top 1% students** in the Indian School Certificate Examination conducted by [CISCE](#).
- **Putani Vignana International Science Talent Examination, 2010** | **Ranked third nationwide**. [[certificate](#)]

SERVICES AND TEACHING ASSISTANTSHIPS

- **Reviewing**
 2026: NeurIPS
 2025: AAAI, NeurIPS
 2024: ICML, ACL, ICML Mechanistic Interpretability Workshop
 2023: NeurIPS R0-FoMo Workshop
- **Volunteering**
 2024: Review of Application Materials Program (RAMP)
 2022: NeurIPS Volunteer
 2021: ACL Volunteer
- **Modern Natural Language Processing (CS-552)**
Master's-level course covering the foundations of natural language processing (NLP), neural language modeling, transformers, and large language models (LLMs).
 - **Head Teaching Assistant (2026):** Led course operations, coordinating PhD teaching assistants and Master's student assistants; managed course communication and forum announcements; resolved ad hoc course-related issues; led exercise sessions and delivered lectures; designed and graded course exams; and co-designed the course project.
 - **Teaching Assistant (2025):** Led exercise sessions and contributed to designing and grading exams.
- **Introduction to Programming (CS-107)**
Bachelor's-level course on programming fundamentals and object-oriented programming in Java.
 - **Teaching Assistant (2025):** Coordinated Bachelor's student assistants, led exercise sessions, answered student queries, and designed and graded the course project and exams.
- **Logic in Computer Science (CS F214)**
Bachelor's-level introductory course on logic, covering first-order logic, model checking, and program verification.
 - **Teaching Assistant (2019):** Supported students with coursework, assignments, and conceptual understanding of course material.
- **Student Faculty Council (SFC) - Member, 2018 - 2019**
 Discussed the academic concerns of students with the faculty members and the HoD, Department of Mathematics.